

Schema Calculus: Piping and Hiding

Example 1 : Schema piping (output - to - input pipeline)

<i>Send</i>	
$chan! : \mathbb{N}$	
$chan! > 0$	

<i>Receive</i>	
$chan? : \mathbb{N}$	
$chan? < 100$	

$$Pipeline \hat{=} Send \gg Receive$$

Example 2 : Schema hiding

<i>State</i>	
$x : \mathbb{N}$	
$y : \mathbb{N}$	
$temp : \mathbb{N}$	
$x < y$	
$temp > 0$	

$$Visible \hat{=} State \setminus (temp)$$

Example 3 : Combined piping and hiding

<i>OpA</i>	
$x : \mathbb{N}$	
$x' : \mathbb{N}$	
$x' > x$	

<i>OpB</i>	
$x : \mathbb{N}$	
$x' : \mathbb{N}$	
$x' < 100$	

$$OpFiltered \hat{=} OpA \mathbin{\text{;}} OpB \setminus (x)$$